

Last Time

Sampling Distribution of $\bar{X}$

The Central Limit Theorem

Confidence Intervals

When σ^2 is Unknown

Other Confidence Intervals

Today

Confidence Intervals

When σ^2 is Unknown

Other Confidence Intervals: Variance

Hypothesis Testing: Single Means

Research vs. Statistical Questions

The Null Hypothesis

The p -value

The Z-test

Test of Proportions

Back to the Normal Distribution

Consider a standard normal Z random variable.

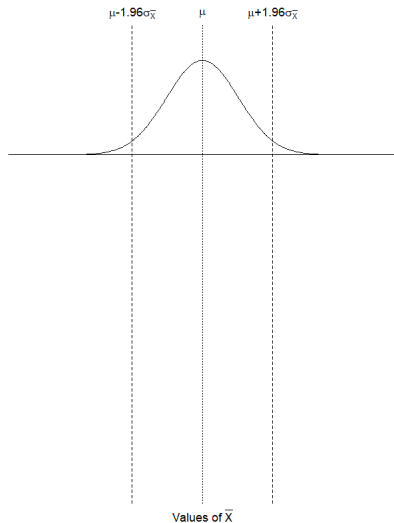
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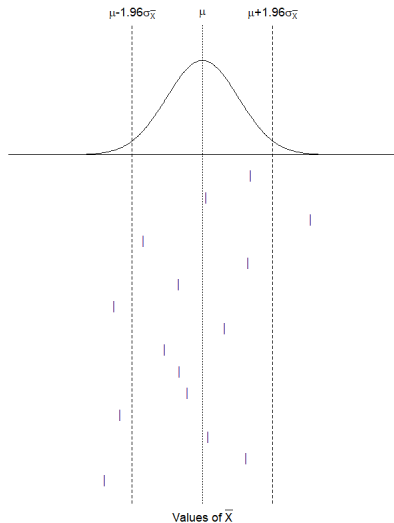
Place upper and lower bounds symmetrically around $\mu_Z = 0$.

Suppose we want the probability of sampling a z -score between those boundaries to be 95%. What are the values of the boundaries?

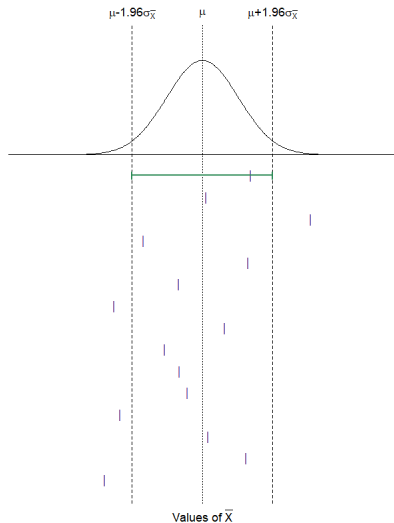
Understanding Confidence Intervals



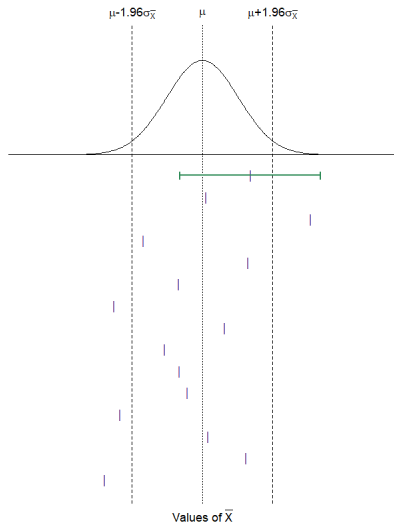
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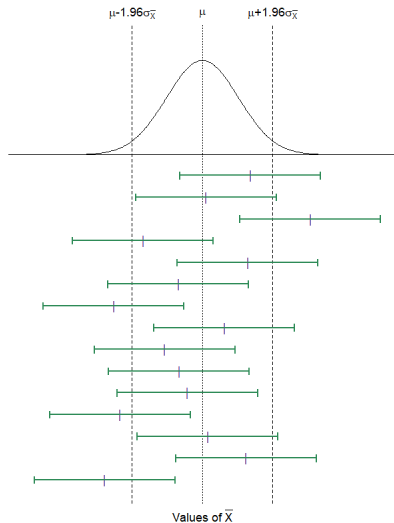
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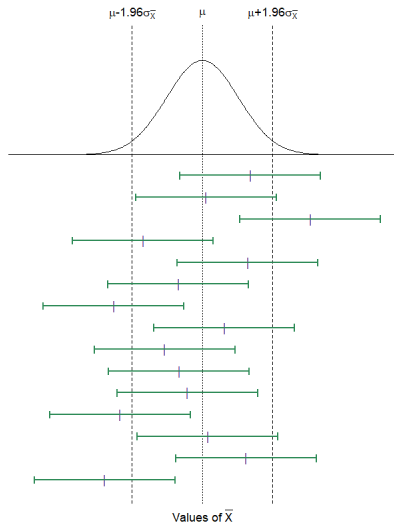


What Is Random?

The interval $\bar{X} \pm 1.96\sigma_{\bar{X}}$ will cover μ for 95% of the values of $\bar{X}$ that we observe.
What is random?

- ▶ $\bar{X}$
- ▶ σ
- ▶ μ
- ▶ Interval $[L, U]$
- ▶ Interval width $U - L$

Understanding Confidence Intervals



Another Way To Think About It

How do we go from an interval around $Z_{\bar{X}}$ to an interval around $\bar{X}$?
Start with what we know:

$$\begin{aligned} Pr(-1.96 < Z_{\bar{X}} < 1.96) \\ = .95 \end{aligned}$$

Another Way To Think About It

How do we go from an interval around $Z_{\bar{X}}$ to an interval around $\bar{X}$?
Start with what we know:

$$\begin{aligned} & Pr(-1.96 < Z_{\bar{X}} < 1.96) \\ &= .95 \\ &= Pr\left(-1.96 < \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} < 1.96\right) \\ &= Pr(-1.96\sigma_{\bar{X}} < \bar{X} - \mu < 1.96\sigma_{\bar{X}}) \\ &= Pr(-\bar{X} - 1.96\sigma_{\bar{X}} < -\mu < -\bar{X} + 1.96\sigma_{\bar{X}}) \\ &= Pr(\bar{X} + 1.96\sigma_{\bar{X}} > \mu > \bar{X} - 1.96\sigma_{\bar{X}}) \\ &= Pr(\bar{X} - 1.96\sigma_{\bar{X}} < \mu < \bar{X} + 1.96\sigma_{\bar{X}}) \end{aligned}$$

What Is Random

$$Pr(\bar{X} - 1.96\sigma_{\bar{X}} < \mu < \bar{X} + 1.96\sigma_{\bar{X}}) = 0.95$$

Suppose $\bar{X} = 97$ and $\sigma_{\bar{X}} = 4$. Then

$$\bar{X} - 1.96\sigma_{\bar{X}} = 89.16$$

$$\bar{X} + 1.96\sigma_{\bar{X}} = 104.84$$

Do we write $Pr(89.16 < \mu < 104.84) = 0.95$?

APA Publication Manual:

Write “ $\bar{X} = 97$, 95% *CI* [89.16, 104.84]”

Back up

Data: $X \sim \mathcal{N}(\mu, \sigma^2)$

If $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/N)$, then

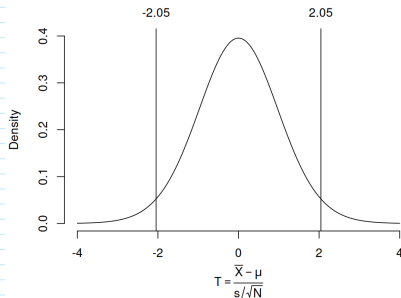
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{N}} \sim \mathcal{N}(0, 1).$$

So $\pm 1.96\sigma_{\bar{X}} = \pm 1.96\sigma/\sqrt{N}$ determines the width of the 95% confidence interval around $\bar{X}$.

Suppose σ^2 is Unknown

Assume $X \sim \mathcal{N}(\mu, \sigma^2)$.¹

$$T = \frac{\bar{X} - \mu}{s/\sqrt{N}} \sim t(N - 1)$$



¹We need this assumption because we need the statistics $\bar{X}$ and s^2 to be independent. This is only true if X is normally distributed.

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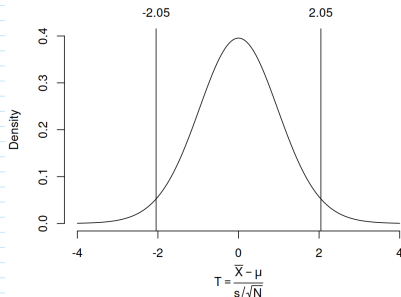
$$T = \frac{\bar{X} - \mu}{s/\sqrt{N}} \sim t(N-1)$$

Z-interval:

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{N}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{N}} \right]$$

T-interval:

$$\left[\bar{X} - t_{\alpha/2} \frac{s}{\sqrt{N}}, \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{N}} \right]$$



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X Not Normally Distributed

```
# Draw 10000 samples of size 30 from an exponential distribution with mean 5 and compute the sample means
bignum <- 10000 # Number of samples
N <- 30         # Sample size
lambda <- 5     # Population mean
conf <- .95     # Confidence level

# Sample from an Exp(lambda) distribution
# To change this to a normal, use rnorm(lambda,lambda) so the mean and variance is the same as the exponential
x <- matrix(rexp(30*10000,1/lambda),ncol=30)

# Compute the means and standard errors for each sample
means <- rowMeans(x)
ses <- apply(x,1,sd)/sqrt(N) # Sample sd divided by sqrt(N)

# Variance known: sigma^2 = 25
z.limits <- qnorm(c((1-conf)/2,1-(1-conf)/2))
z.lower <- means + z.limits[1]*lambda/sqrt(N)
z.upper <- means + z.limits[2]*lambda/sqrt(N)

# Variance unknown: use s^2
t.limits <- qt(c((1-conf)/2,1-(1-conf)/2),N-1)
t.lower <- means + t.limits[1]*ses
t.upper <- means + t.limits[2]*ses

# Compute the probabilities that the z and t intervals
# contain lambda
conf.z <- mean(lambda >= z.lower & lambda <= z.upper)
conf.t <- mean(lambda >= t.lower & lambda <= t.upper)

rbind(c("Confidence","Sigma known","Sigma unknown"),
      c(0.95,round(conf.z,2),round(conf.t,2)))
```

Remember the χ^2 Distribution

Assume again $X \sim \mathcal{N}(\mu, \sigma^2)$, so $\frac{X - \bar{X}}{\sigma} \sim \mathcal{N}(0, 1)$ if N is large enough that $\bar{X} \approx \mu$.
Then

$$s^2 = \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N - 1}$$

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Then

$$s^2 = \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N - 1}$$
$$\rightarrow \frac{(N - 1)s^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^N (X_i - \bar{X})^2 = \sum_{i=1}^N \frac{(X_i - \bar{X})^2}{\sigma^2} = \sum_{i=1}^N \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \sum_{i=1}^N Z_i^2,$$

We take away a degree of freedom because we substituted $\bar{X}$ for μ . So

$$\frac{(N - 1)s^2}{\sigma^2} \sim \chi^2(N - 1).$$

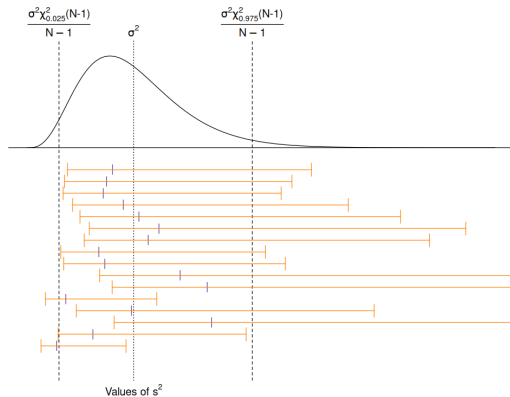
Confidence Interval Around the Sample Variance

$$\begin{aligned} Pr(\chi^2_{.025}(N-1) \leq \frac{(N-1)s^2}{\sigma^2} \leq \chi^2_{.975}(N-1)) \\ = 0.95 \end{aligned}$$

Confidence Interval Around the Sample Variance

$$\begin{aligned} & Pr(\chi_{.025}^2(N-1) \leq \frac{(N-1)s^2}{\sigma^2} \leq \chi_{.975}^2(N-1)) \\ &= 0.95 \\ &= Pr\left(\frac{1}{\chi_{.975}^2(N-1)} \leq \frac{\sigma^2}{(N-1)s^2} \leq \frac{1}{\chi_{.025}^2(N-1)}\right) \\ &= Pr\left(\frac{(N-1)s^2}{\chi_{.975}^2(N-1)} \leq \sigma^2 \leq \frac{(N-1)s^2}{\chi_{.025}^2(N-1)}\right). \end{aligned}$$

Confidence Interval Around the Sample Variance²



²I'm skipping some important steps.

GSS Data

In 1980 the ~~median~~ mean age of the US population was 30 years old.
For the GSS data, $\bar{X} = 39.25$ and $s^2 = 196.19$, $N = 873$.

$$\bar{X} \pm t_{.025}(N-1)s/\sqrt{N} = 39.25 \pm 1.96(14.01/29.55) = [39.21, 39.28]$$

$$\left[\frac{(N-1)s^2}{\chi_{.975}^2(N-1)}, \frac{(N-1)s^2}{\chi_{.025}^2(N-1)} \right] = \left[\frac{(872)(196.19)}{955.73}, \frac{(872)(196.19)}{792.06} \right] = [179.00, 215.99]$$

GSS Data

```
### Confidence intervals for the GSS Age data
library("DescTools")

data <- read.csv("GSS_1980.csv",header=TRUE)
age <- data$AGE

N <- length(age)
mean.age <- mean(age)
sd.age <- sd(age)

mean.limits <- mean.age + qt(c(.025,.975),N-1)*sd.age/N
var.limits <- sd.age^2*(N-1)/qchisq(c(.975,.025),N-1)

cat("Mean age = ",
    mean.age," , 95% CI [",
    mean.limits[1]," ,",mean.limits[2],"]\n",
    "Age variance= ",
    sd.age^2," , 95% CI [",
    var.limits[1]," ,",var.limits[2],"]")

# See also:
MeanCI(age)
VarCI(age)
```

Z- and t -scores

- ▶ Both Z and t express a variable's **distance from its mean** in units of **standard deviations**.
- ▶ For some variable V (X , $\bar{X}$, etc.),

$$z_V = \frac{V - \mu_V}{\sigma_V}$$

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- ▶ The farther a value of a measurement V is from the mean of its distribution (μ_V), the smaller the probability of observing measurements as extreme as that value (Chebyshev's Inequality).

Tables

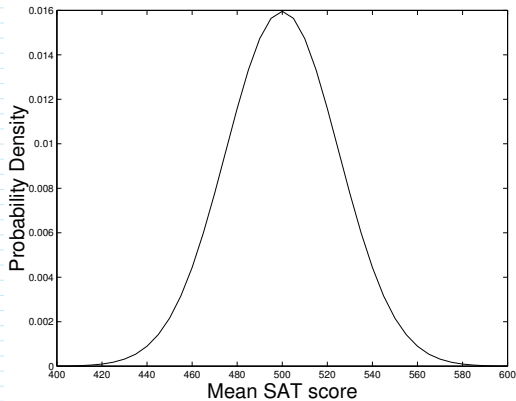
Download these tables in case of dystopian future:

`pnorm()` Normal table

`pt(..., $\nu$ )` T table

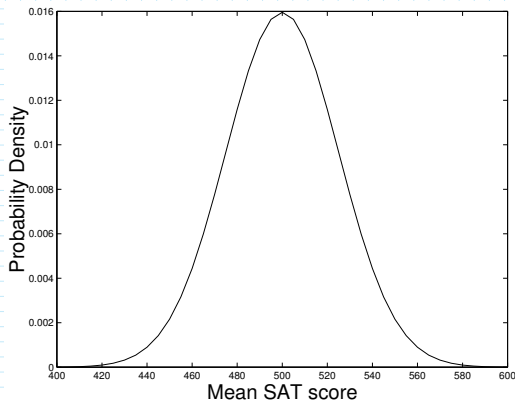
The Logic of Hypothesis Testing

A sample $\{X_1, X_2, \dots, X_{16}\}$ is drawn from a distribution with mean $\mu = 500$ and $\sigma = 100$.



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What is the probability of obtaining a sample mean $\bar{X}$ *as extreme or more extreme* than the one we computed for our sample ($\bar{x}$) given that this is what the sampling distribution of the mean must be?

Hypothesis Testing

- ▶ Hypothesis testing is a procedure for deciding if an hypothesis about a population parameter (like μ) is true given the value of the statistic that we compute from a sample.
- ▶ The result of an hypothesis test is a *probablistic* answer to our question: we can compute probabilities associated with our statistical value if the hypothesis is true.
- ▶ Statistical questions are not research questions. A “significant” hypothesis test does not imply an important or interesting research question.

Research vs. statistical questions

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- ▶ Research question example: on standardized exams, should you answer with the first response that comes to mind, or is it okay to go back and change your answers?
- ▶ Example: Consider a standardized test of academic achievement administered to 5th graders. The national average on this test is $\mu = 200$ and the national standard deviation is $\sigma = 48$.
- ▶ A class of $N = 64$ 5th graders at Indianola Elementary is instructed to give the first answers that come to mind. The average performance was $\bar{X} = 220$.
- ▶ Statistical question: is $\bar{X} = 220$ significantly greater than 200?

The Null Hypothesis

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- ▶ Alternatively, the treatment had the desired effect (improved performance).

$$H_1 : \quad \mu > 200$$

One- vs. Two-Tailed Tests

- ▶ The null hypothesis H_0 always contains the uninteresting result (an “=” sign):

$$H_0 : \mu = \mu_0$$

- ▶ Two-tailed test:

$$H_1 : \mu \neq \mu_0$$

- ▶ One-tailed test:

$$H_1 : \mu > \mu_0 \text{ OR } H_1 : \mu < \mu_0$$

- ▶ The scientific question determines the direction of the alternative.
- ▶ Two-tailed tests are rarely interesting.

The p -value

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The p -value

- ▶ When the sample mean we observe ($\bar{x}$) is highly unlikely when the null hypothesis is true, we will *reject* H_0 .
- ▶ BUT what does “highly unlikely” mean?
 - ▶ We must select a criterion probability against which to evaluate $Pr(\bar{X} \geq \bar{x} \mid \mu = \mu_0)$.
 - ▶ This criterion is called the α -level and is traditionally set to 0.05 (0.005).
 - ▶ The α -level is the probability of making a mistake (Type I error).

The α -Level

- ▶ We need to set rules about when H_0 will be *rejected* so that our probability of making a mistake is
 - ▶ consistent (fixed) from experiment to experiment, and
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- ▶ Significance level = $\alpha = P(\text{Type I error} \mid H_0 \text{ true})$

Errors in Hypothesis Testing

How many ways can we make a mistake in our statistical conclusion?

Errors in Hypothesis Testing

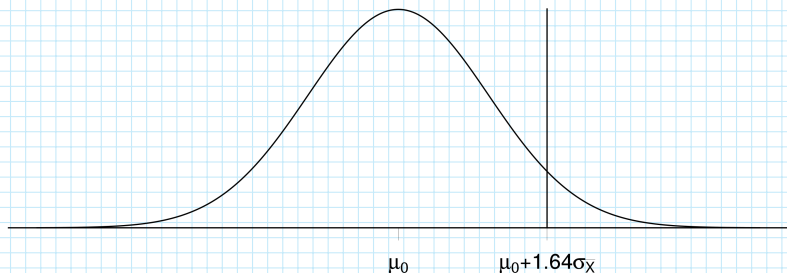
How many ways can we make a mistake in our statistical conclusion?

Two ways:

State of the world	Decision	
	Fail to reject H_0	Reject H_0
H_0 True	Correct	Type I Error
H_0 False	Type II Error	Correct

Incorrectly Rejecting H_0

We expect to see extreme, unlikely means about $100\alpha\%$ of the time.



Values of $\bar{X}$ if H_0 is true

If we reject H_0 incorrectly, we have committed a Type I error.

The Z-test

- ▶ Testing hypotheses about single means *when σ is known*.
- ▶ Achievement Test: For $N = 64$, we observe $\bar{X} = 220$. The national average of these scores is $\mu = 200$ and the standard deviation is $\sigma = 48$.

Three Ways to Do A Z-test

1. **Compare the p -value of $\bar{X}$ to α -level.**
2. Compare the z -score of $\bar{X}$ to the z -score associated with the desired α -level.
3. Compare $\bar{X}$ to the value of the mean that has the z -score associated with the desired α -level.

Use Z-score method

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- ▶ OR: Determine the *critical values* of the test statistic as determined by the α -level.
- ▶ Compare the test statistic to the critical values. If the test statistic is
 - ▶ more extreme than the critical values, reject H_0 .
 - ▶ not as extreme as the critical values, fail to reject H_0 .

Test of Proportions

Let $\{X_1, X_2, \dots, X_N\}$ be a sequence of (0,1) Bernoulli(p) observations, where p is the population probability of success. Then $\bar{X} = \hat{p}$, the proportion of “successes” in the sample and an estimate of p .

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We know that

$$\begin{aligned} E(X) = \mu &= p \text{ and} \\ \text{Var}(X) = \sigma^2 &= p(1 - p). \end{aligned}$$

So $E(\bar{X}) = E(X) = p$ and $\text{Var}(\bar{X}) = \sigma^2/N = p(1 - p)/N$.

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So $E(\bar{X}) = E(X) = p$ and $\text{Var}(\bar{X}) = \sigma^2/N = p(1 - p)/N$.

Central Limit Theorem: for large enough N , $\bar{X} = \hat{p} \sim \mathcal{N}(p, p(1 - p)/N)$.

This is one of those rare circumstances where we know σ^2 : the null hypothesis $H_0 : p = p_0$ determines both the mean and the variance of the sampling distribution of $\bar{X} = \hat{p}$.

Birth Order Effects

$$H_0 : p \leq 0.5$$

$$H_1 : p > 0.5$$

Set $\alpha = 0.05$.

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Set $\alpha = 0.05$.

$$\bar{X} = 10/15 = 0.6667$$

$$\sigma_{\bar{X}}^2 = 0.5(1 - 0.5)/15 = 0.0167$$

$$Z_{\bar{X}} = \frac{0.6667 - 0.50}{\sqrt{0.0167}} = 1.2910$$

$$Pr(Z \geq 1.2910) = 0.0984$$

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Birth Order Effects

```
prop.test(10,15,p=.5,alternative="greater")
```

1-sample proportions test with continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.0667, df = 1, p-value = 0.1508

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4234848 1.0000000

sample estimates:

p

0.6666667

Remember the Correction For Continuity

Compare:

```
prop.test(10,15,p=.5,alternative="greater",correct=FALSE)
```

1-sample proportions test without continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.6667, df = 1, p-value = 0.09835

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4551721 1.0000000

sample estimates:

p

0.6666667

Yates Correction

We computed $Pr(\hat{p} > 0.6667 \mid p_0 = .5) = Pr(Z > 1.2910) = 0.0984$.

But $\hat{p} = 0.6667$ was computed as $\frac{10}{15}$. Because $\bar{X} = \hat{p}$ is discrete,

$$Pr(\hat{p} \geq 10/15 \mid p_0 = 0.5) = Pr(\hat{p} > 9.5/15 \mid p_0 = 0.5).$$

The *real lower limit* of 10 is 9.5, which is where we need to consider the starting point to compute the area under the (continuous) normal density function.

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The *real lower limit* of 10 is 9.5, which is where we need to consider the starting point to compute the area under the (continuous) normal density function.

Recompute:

$$\hat{p} = 9.5/15 = 0.6333, \text{ so}$$

$$Z_{\hat{p}} = \frac{0.6333 - 0.5}{\sqrt{0.0167}} = 1.0318$$

$$Pr(Z \geq 1.0318) = 0.1508$$

Exact Binomial Test

The p -value with the Yates correction is 0.1508.

Compare this to the exact binomial test:

$$Pr(X \geq 10 \mid p = 0.5) = 0.1509$$

```
binom.test(10,15,alternative="greater")
```

```
Exact binomial test
```

```
data: 10 and 15
```

```
number of successes = 10, number of trials = 15, p-value = 0.1509
```

```
alternative hypothesis: true probability of success is greater than 0.5
```

```
95 percent confidence interval:
```

```
0.4225563 1.0000000
```

```
sample estimates:
```

```
probability of success
```

```
0.6666667
```